Shared mathematical structures

The applications repeatedly use differences, energies and fluxes, nonnegative evolution, projection, and coordinate transformations. The projection formula below assumes that the v_i form a nonzero orthogonal basis of the target subspace.

$$\text{difference } A, \qquad \text{energy } A^T C A, \qquad u_{k+1} = A u_k,$$

$$\operatorname{proj}_U b = \sum_i \frac{\langle v_i, b \rangle}{\langle v_i, v_i \rangle} v_i.$$

A model must specify entry units and signs, the assumptions behind its equations, and the distinction between exact invertibility, physical interpretation, and numerical reliability.

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Oriented graphs and incidence matrices

Let the nodes be $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$. Choose an orientation for each edge. For a graph without self-loops, define $A \in \mathbb{R}^{m \times n}$ by

$$a_{ij} = \begin{cases} -1, & e_i \text{ leaves } v_j, \\ +1, & e_i \text{ enters } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Each edge row sums to zero. For a node-potential vector x,

$$(Ax)_i = x_{\text{head}} - x_{\text{tail}}.$$

Thus A is a discrete difference operator. Reversing an edge negates its row and difference, without changing the squared difference.

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Kernel and rank of a connected graph

Connected means that the underlying undirected graph is connected after ignoring the chosen edge directions; strong connectivity is not required. All edge differences vanish exactly when all node potentials agree.

$$N(A) = \operatorname{span}\{\mathbf{1}\}, \qquad \dim N(A) = 1.$$

Each edge row sums to zero, so $A\mathbf{1} = 0$ and the rank is less than n. This does not assert an equal-coefficient dependence among all rows. Equality of potentials along paths proves that the kernel consists exactly of constant vectors. Rank-nullity then gives rank $n - 1$; alternatively use the $n - 1$ independent edge rows of a spanning tree. Thus

$$\operatorname{rank} A = n - 1, \qquad \dim N(A^T) = m - n + 1.$$

The column space $C(A) \subseteq \mathbb{R}^m$ describes attainable edge potential differences. The row space lies in node space and is the orthogonal complement of the constant vectors. The left nullspace describes cycle flows. These dimensions encode the graph topology.

Cycles, flows, and Kirchhoff laws

The equation $Ax = b$ is solvable exactly when $b \in C(A)$, equivalently when every cycle flow $j \in N(A^T)$ satisfies

$$j^T b = 0.$$

Thus signed potential differences sum to zero around every closed loop, the linear-algebra form of Kirchhoff's voltage law.

The left nullspace consists of edge flows satisfying

$$A^T j = 0.$$

At every node, inflow minus outflow must balance, giving Kirchhoff's current law. Fix a spanning tree. Each nontree edge and the unique tree path between its endpoints form a fundamental cycle. Give its edges coefficients +1, −1, or 0 according to orientation; then $A^T j = 0$. Each such cycle contains its own nontree edge and no other nontree edge, so comparison of these coordinates proves independence. There are $m - n + 1$ cycles, equal to the left nullity, so they form a cycle-space basis. Arbitrarily chosen small cycles need not be independent or span the cycle space.

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Cycles, flows, and Kirchhoff laws (continued)

For resistive networks, fix $e = Ax$ as head potential minus tail potential, and $y = Ce$ as weighted potential differences. Physical current j along the chosen arrows satisfies Ohm's law $j = -y$. Positive f denotes external injection into a node. Conservation is $-A^T j = f$, hence $A^T y = f$. Positive y must not simultaneously be called current along an arrow pointing from lower to higher potential.

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Laplacians and network equations

Put positive conductances c_e in diagonal C and define

$$L = A^T C A.$$

For potentials x,

$$x^T L x = (Ax)^T C (Ax) = \sum_e c_e (\Delta x_e)^2 \geq 0.$$

Thus L is symmetric positive semidefinite. For a connected graph with positive conductances, its kernel consists of constant potentials. Without a reference potential, every solution admits an arbitrary common offset.

Ground a node, for example $x_n = 0$, and delete its row and column to obtain L_{red}. Connectivity makes the reduced matrix positive definite.

The network relations are

$$e = Ax, \qquad y = Ce, \quad j = -y, \qquad f = -A^T j = A^T y,$$

and combine into

Laplacians and network equations (continued)

$$A^T C A x = f.$$

Reversing edge directions does not change L. Let each orientation change be $s_e \in \{-1, 1\}$, and set

$$D = \mathrm{diag}(s_1, \ldots, s_m), \qquad A' = DA.$$

Since $D^T = D$, $D^2 = I$, and C is diagonal,

$$(A')^T C A' = A^T D\, C\, DA = A^T C A.$$

The corresponding edge quantities change consistently as $e' = De$, $y' = Dy$, giving

$$(A')^T y' = A^T D^2 y = A^T y = f.$$

Orientation changes only the signed representation of edge differences and currents. It preserves the Laplacian, energy, physical flow, and node balance.

Also,

$$\mathbf{1}^T f = \mathbf{1}^T A^T C A x = (A\mathbf{1})^T C A x = 0,$$

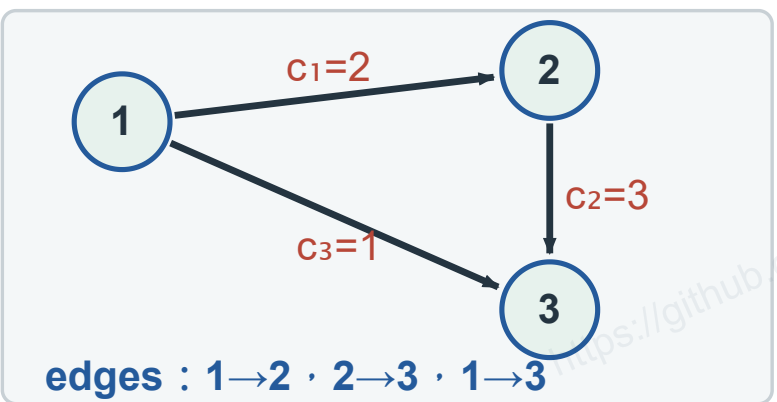
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Laplacians and network equations (continued)

so without an external cross-network source, injections must sum to zero. A nonzero total is a model or data inconsistency that changing the solver cannot repair.

The triangle uses edges $1 \to 2, 2 \to 3, 1 \to 3$, with each incidence row computing head potential minus tail potential.

Triangle network: incidence, orientation, and differences



Rows of A

[-1, 1, 0]

[0,-1, 1]

[-1, 0, 1]

x=(1,4,7)ᵀ

e_1=4-1=3

e_2=7-4=3

e_3=7-1=6

e=Ax=(3,3,6)ᵀ

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A weighted triangular network

Fix edges $1 \to 2, 2 \to 3, 1 \to 3$, with conductances $2, 3, 1$. Each incidence row represents one oriented edge.

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix}.$$

For $e_i = (Ax)_i = x_{\text{head}} - x_{\text{tail}}$ and $x = (1, 4, 7)^T$, the rows give

$$Ax = \begin{bmatrix} 4 - 1 \\ 7 - 4 \\ 7 - 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} = e.$$

Multiply by $C = \text{diag}(2, 3, 1)$.

$$y = Ce = (6, 9, 6)^T.$$

The physical currents along the arrows are $j = (-6, -9, -6)^T$, all flowing in reverse. Multiplication by A^T gives the external injections at nodes $1, 2, 3$.

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A weighted triangular network (continued)

$$f = A^T y = \begin{bmatrix} -6 - 6 \\ 6 - 9 \\ 9 + 6 \end{bmatrix} = \begin{bmatrix} -12 \\ -3 \\ 15 \end{bmatrix}.$$

Their sum is $-12 - 3 + 15 = 0$, as conservation requires. The weighted Laplacian is

$$L = A^T C A = \begin{bmatrix} 3 & -2 & -1 \\ -2 & 5 & -3 \\ -1 & -3 & 4 \end{bmatrix}, \qquad Lx = f, \quad L\mathbf{1} = 0.$$

The incidence rank is 2 and its kernel consists of constants. Grounding node 1 at zero and deleting its row and column gives $x_2 = 3, x_3 = 6$. Adding the common offset 1 recovers $x = (1, 4, 7)^T$. Changing the reference potential preserves differences and flows.

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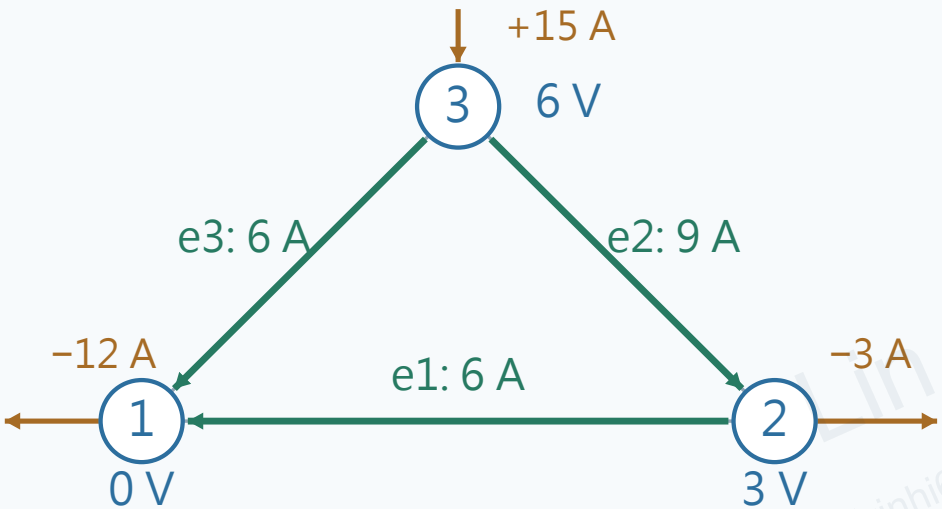
Actual currents, signed coefficients, and power

Shift all potentials by -1 V to ground node 1: $x=(0,3,6)^{\mathsf{T}}$ V. Currents remain unchanged; $c=(2,3,1)$ is measured in siemens.

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Actual currents, signed coefficients, and power (continued)

Actual currents after grounding node 1



Green: physical current; amber: external injection.

Signs, units, and power

$x = (0,3,6)$ V

$e = Ax = (3,3,6)$ V

$y = Ce = (6,9,6)$ A

$j = -y = (-6,-9,-6)$ A

$f = -A^{T}j = (-12,-3,15)$ A

$P = x^{T}Lx = 18+27+36$

$= 81$ W

The quadratic $x^{T}Lx$ has units V · A=W: it is dissipated power. Multiplying by ½ gives an optimization quadratic, not energy stored in ideal resistors.

Solve the grounded system explicitly

With node 1 grounded, $x_1=0$. The other two equations are

$$\begin{bmatrix} 5 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 15 \end{bmatrix}.$$

The determinant is 11, so

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -3 \\ 15 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 33 \\ 66 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Check the omitted first row as well: $-2x_2-x_3=-6-6=-12=f_1$. All three original equations must hold to verify conservation.

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Grounding cannot replace conservation

If the data are mistakenly $\tilde{f} = (-11, -3, 15)^T$ A, the reduced right-hand side remains (−3,15)ᵀ, so the solver still returns x=(0,3,6)ᵀ V. But

$$Lx - \tilde{f} = (-1, 0, 0)^T \text{ A}, \qquad \mathbf{1}^T \tilde{f} = 1 \text{ A} \neq 0.$$

The full system is inconsistent because 1ᵀL=0. A reference potential alone cannot hide the missing 1 A balance. If an ideal voltage source is physically attached to the grounded terminal, its current is an unknown reaction that must be modeled; it does not make all three prescribed injections consistent.

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Springs and energy forms

Let u denote displacements and $e = Au$ spring elongations. The constitutive relation is

$$y = Ce, \qquad C = \operatorname{diag}(c_i).$$

Equilibrium gives

$$A^T y = f.$$

Combining gives

$$A^T C A u = f.$$

The stiffness matrix is symmetric positive semidefinite for positive spring constants. It is positive definite when the constraints remove all zero-energy motions. The strain energy is

$$\frac{1}{2} u^T K u = \frac{1}{2} \sum_i c_i e_i^2.$$

For a free connected one-dimensional chain, constant translation stretches no spring and lies in the nullspace. Boundary constraints or a displacement reference remove this mode.

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Springs and energy forms (continued)

Finite-element assembly uses the same pattern, adding local matrices $A_e^T C_e A_e$ into a global stiffness matrix.

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Second differences and boundaries

For N interior nodes with both endpoints fixed, the $(N+1) \times N$ difference matrix maps u to $(u_1, u_2 - u_1, \ldots, u_N - u_{N-1}, -u_N)^T$. It gives

$$K = A^T A = \begin{bmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \ddots \\ 0 & -1 & 2 & \ddots \end{bmatrix}.$$

Before dividing by h^2,

$$(Ku)_i = 2u_i - u_{i-1} - u_{i+1} \approx -h^2 u''(x_i).$$

Known boundary values move to the right-hand side. A free right endpoint removes the last spring contribution, changing the final diagonal entry. Boundary conditions determine the matrix itself.

A free-free chain has a constant-displacement nullspace and singular stiffness. The force vector must balance the rigid mode; a reference displacement can then select a representative, with reaction forces included where constraints impose them.

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Engineering model checks

Before solving $Ku = f$, identify the following.

1. Degrees of freedom and units.
2. Displacement, force, and elongation signs.
3. difference matrix A;
4. constitutive diagonal C;
5. Boundary constraints and grounded modes.
6. Symmetry and positive definiteness.
7. Nonnegative energy and rigid modes.
8. Conditioning under mesh refinement.

A missing boundary condition often appears as a zero eigenvalue and translation mode. Thin elements or large coefficient contrasts can also cause poor conditioning in an invertible positive-definite system.

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Markov matrices

A column-stochastic matrix satisfies

$$P_{ij} \geq 0, \qquad \sum_i P_{ij} = 1.$$

For a probability vector u_k,

$$u_{k+1} = Pu_k,$$

we have

$$\mathbf{1}^T u_{k+1} = \mathbf{1}^T u_k.$$

so total probability remains 1. Also,

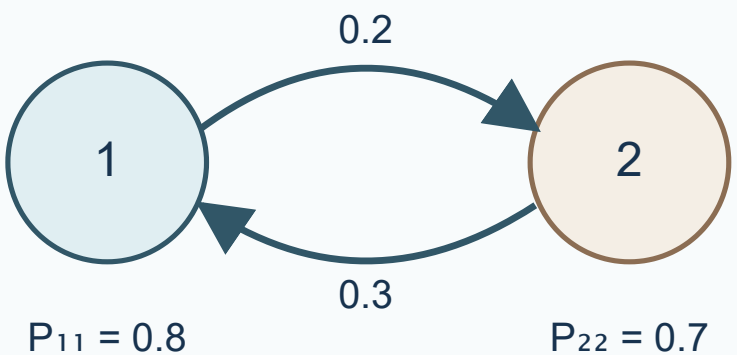
$$\mathbf{1}^T P = \mathbf{1}^T,$$

Thus 1 is a left eigenvalue, and hence an eigenvalue of P, since transposition preserves the characteristic polynomial. In general $P\mathbf{1} \neq \mathbf{1}$. A stationary distribution is a right eigenvector satisfying

$$Pu_\infty = u_\infty.$$

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Two-state Markov chain: transitions and convergence



$u_0 = (0.5, 0.5)$

$u_1 = (0.55, 0.45)$

$u_2 = (0.575, 0.425)$

$u_3 = (0.5875, 0.4125)$

$u_\infty = (0.6, 0.4)$

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Let $P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$, with $u_{k+1} = Pu_k$. Each column is a source and each row a destination; for example $P_{21} = 0.2$ is the probability of moving from state 1 to state 2. Here $u_1 = (11/20, 9/20)^T$ and $\pi = (3/5, 2/5)^T$ is stationary.

The other eigenvalue is $1/2$, hence $u_k = \pi + (1/2)^k(u_0 - \pi)$: the deviation halves each step.

Convergence and spectral radius

If every entry of P is positive, let $\alpha = n \min_{ij} P_{ij} > 0$. For $\alpha < 1$, write $P = \alpha \mathbf{1}\mathbf{1}^T/n + (1-\alpha)R$, with column-stochastic R. For probability vectors u, v, $\|Pu - Pv\|_1 \le (1-\alpha)\|u - v\|_1$. The probability simplex is complete in ℓ_1, so the contraction theorem gives a unique stationary vector and geometric convergence. At $\alpha = 1$, one step produces the uniform distribution. Every stationary component is positive.

With zero entries allowed, the convergence version of Perron–Frobenius requires irreducibility and aperiodicity. These imply a unique positive stationary distribution attracting every initial probability vector. The periodic matrix $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has $P^2 = I$: for example $(1,0)^T$ and $(0,1)^T$ alternate without convergence, whereas the stationary initial distribution $(1/2, 1/2)^T$ stays fixed. For reducible $P = I$, every standard basis vector is stationary, and the limit depends on the initial state.

If P is diagonalizable, order the eigenvalues as

$$1 = \lambda_1 > |\lambda_2| \ge \cdots,$$

Then

Convergence and spectral radius (continued)

$$u_k = P^k u_0 = \sum_i c_i \lambda_i^k v_i.$$

The second eigenvalue controls mixing through

$$\|u_k - u_\infty\| \leq C|\lambda_2|^k.$$

For example,

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

The eigenvalues are $1, 0.5$, with stationary vector $u_\infty = (0.6, 0.4)^T$. The nonstationary mode halves each step. A nondiagonalizable matrix adds Jordan factors k^{m-1}; spectral radius still controls convergence, but a pure $C|\lambda_2|^k$ bound needs modification.

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Population and input-output models

A Leslie matrix separates age classes.

$$n_{k+1} = Ln_k.$$

Its first row contains fertility rates and its subdiagonal survival rates. For an irreducible, aperiodic nonnegative transition graph, Perron–Frobenius gives a unique positive dominant eigenvalue λ. Normalize its eigenvector to obtain the stable age distribution; λ is the long-run growth factor. Without these hypotheses, periodic behavior or isolated cohorts can prevent a single limiting distribution.

The consumption coefficient A_{ij} is input from sector i needed for one unit of sector j's output. Total output p, less internal consumption Ap, leaves

$$(I - A)p = y.$$

If $\rho(A) < 1$, the Neumann series converges.

$$(I - A)^{-1} = I + A + A^2 + \cdots.$$

For a proof, examine the finite sum.

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Population and input-output models (continued)

$$(I - A)(I + A + \cdots + A^N) = I - A^{N+1}.$$

Since $\rho(A) < 1$ is equivalent to $A^N \to 0$, the sums tend to $(I - A)^{-1}$. For $A \geq 0$, every term is nonnegative, hence so is the inverse. Nonnegative demand y yields

$$p = (I - A)^{-1} y$$

which accumulates direct and indirect production requirements. If some $|\lambda| \geq 1$, the series fails to converge. The matrix $I - A$ might remain invertible, but that alone does not ensure nonnegative production or a sustainable long-run model.

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Linear-program geometry

Standard form:

$$\begin{aligned} &\min\ c^T x, \\ &\text{s.t. } Ax = b, \quad x \geq 0. \end{aligned}$$

For example,

$$\min\ 5x_1 + 3x_2 + 8x_3$$

subject to

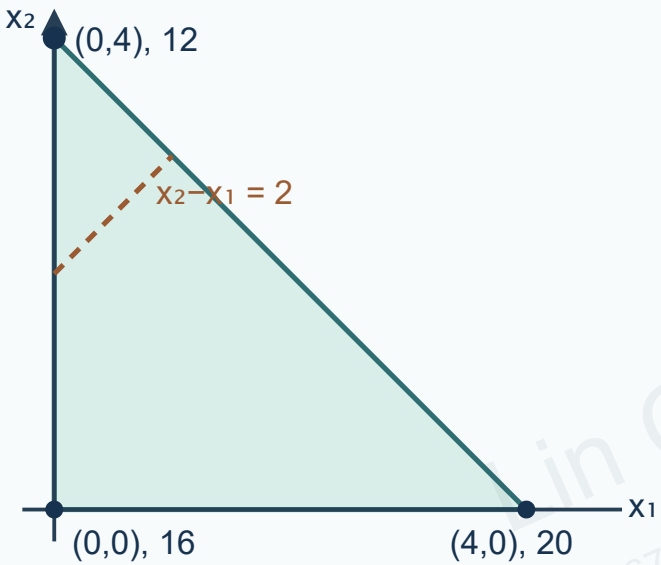
$$x_1 + x_2 + 2x_3 = 4, \qquad x_1, x_2, x_3 \geq 0.$$

The equality is a plane. Nonnegativity cuts out a triangle, and objective level sets are parallel planes. A linear objective on this nonempty compact polytope attains an optimum at a vertex.

Assume redundant equalities have been removed from $A \in \mathbb{R}^{m \times n}$, so its rank is m. A basic feasible solution selects m independent columns as square B, sets nonbasic variables to zero, solves $Bx_B = b$, and requires $x_B \geq 0$.

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Linear programming: flatten the feasible plane



$x_3 = (4 - x_1 - x_2) / 2$

$z = 16 + x_1 - x_2$

$z = 14$ on dashed line

minimum $z = 12$ at $(0,4)$

$x_3 \geq 0 \Leftrightarrow x_1 + x_2 \leq 4$

Eliminate $x_3 = (4 - x_1 - x_2)/2$. Nonnegativity gives the triangle and the objective becomes $z = 16 + x_1 - x_2$. The dashed $z = 14$ line and the optimum $z = 12$ are distinct level lines. Vertex objectives are 16, 20, and 12; the optimizer corresponds to $(x_1, x_2, x_3) = (0, 4, 0)$.

Simplex, duality, and numerical conditions

For basis B, the objective value is

$$c_B^T B^{-1} b.$$

A nonbasic column a_j has reduced cost

$$\bar{c}_j = c_j - c_B^T B^{-1} a_j.$$

If every $\bar{c}_j \geq 0$, the feasible change $x_B \mapsto x_B - B^{-1} a_j t$ changes the objective by $\bar{c}_j t \geq 0$, certifying optimality. For $\bar{c}_j < 0$, let x_j enter and apply the ratio test.

$$t = \min_{i:\, (B^{-1}a_j)_i > 0} \frac{x_{Bi}}{(B^{-1}a_j)_i}$$

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Simplex, duality, and numerical conditions (continued)

Among positive denominators, take the smallest nonnegative ratio so the first basic variable reaching zero leaves. A zero basic value must participate and can give a zero step. If all denominators are nonpositive, the entering variable can grow without bound while decreasing the objective, giving an unboundedness certificate. A nondegenerate pivot reaches an adjacent vertex; a degenerate pivot can leave the point fixed. When both primal and dual are strictly feasible, interior-point methods can approach the optimum along the positive central path below. Without strict feasibility, preprocessing or other feasibility treatment is needed.

The primal and dual are

$$\min c^T x \quad \text{s.t. } Ax = b,\ x \geq 0,$$

$$\max b^T y \quad \text{s.t. } A^T y \leq c.$$

Every feasible pair satisfies weak duality because

$$c^T x - b^T y = x^T (c - A^T y) \geq 0.$$

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Simplex, duality, and numerical conditions (continued)

Equal objective values for feasible primal and dual points certify optimality. For finite-dimensional LP, feasibility and a finite attained optimum on one side imply an equal attained optimum on the other by strong duality. The explicit matching feasible pair in this example already certifies optimality. Optimal solutions satisfy complementary slackness.

$$x_j(c_j - A_j^T y) = 0.$$

A positive variable has zero reduced cost. Dual variables describe marginal demand prices only while the current optimal basis stays valid.

Degeneracy means a zero basic variable and can give multiple bases at one vertex. A tiny pivot is a separate numerical issue. Interior-point methods use

$$Ax = b, \qquad A^T y + s = c, \qquad x_j s_j = \mu > 0$$

for the central path, solve linearized Newton corrections, and let $\mu \downarrow 0$. Report solver status, primal and dual residuals, duality gap, and conditioning alongside the vector.

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A complete pivot and dual certificate

Minimize $5x_1 + 3x_2 + 8x_3$, with $x_1 + x_2 + 2x_3 = 4$, $x \geq 0$. Initially choose x_3 as basic, so $B = [2]$, $c_B = 8$, and

$$x_3 = 2 - \tfrac{1}{2}x_1 - \tfrac{1}{2}x_2, \qquad z = 16 + x_1 - x_2.$$

The reduced cost of x_2 is -1. Its direction denominator is $B^{-1}a_2 = 1/2 > 0$, giving ratio $2/(1/2) = 4$. Increasing x_2 to 4 makes x_3 zero. After the pivot,

$$x_2 = 4 - x_1 - 2x_3, \qquad z = 12 + 2x_1 + 2x_3.$$

all nonbasic coefficients are nonnegative, so $x = (0, 4, 0)^T$, $z = 12$, is optimal. Generally, substitute $x_B = B^{-1}b - \sum_{j \notin B} B^{-1}a_j x_j$ into the objective to get $z = c_B^T B^{-1} b + \sum_{j \notin B} \bar{c}_j x_j$. For $x_j \geq 0, \bar{c}_j \geq 0$, no feasible point has a lower value.

Choose dual $y = 3$. Then $A^T y = (3, 3, 6)^T \leq (5, 3, 8)^T$, with value $4y = 12$. Matching feasible values independently certify optimality. Positive x_2 corresponds to the tight second dual constraint; the other slacks are both 2.

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A zero ratio is a valid degenerate pivot

Minimize $-x_1$, subject to $x_1 + s_1 = 0$, $x_1 + s_2 = 1$, and nonnegativity. The initial basis (s_1, s_2) gives $(x_1, s_1, s_2) = (0, 0, 1)$. Increase x_1 by t.

Constraint	Basic variable as a function of t	Positive denominator	Nonnegative ratio
1	$s_1 = 0 - t$	1	0
2	$s_2 = 1 - t$	1	1

The minimum ratio is 0. Variable s_1 leaves and x_1 enters, but the point stays fixed. Excluding zero and taking $t = 1$ gives $s_1 = -1$, violating feasibility.

The new dictionary is $x_1 = -s_1$, $s_2 = 1 + s_1$, with objective $s_1 \geq 0$. Thus value 0 is optimal. Degenerate pivots may not improve the objective and can cycle. Bland's rule chooses the smallest-index improving entering variable, then the smallest-index basic variable among minimum-ratio ties. It avoids cycling without discarding valid zero ratios.

Functions as vectors

For real functions in $L^2([0, 2\pi])$, define

$$\langle f, g \rangle = \int_0^{2\pi} f(x) g(x)\, dx.$$

For complex functions conjugate the first argument. Strictly, L^2 consists of equivalence classes modulo equality almost everywhere; otherwise a function nonzero at only one point could have zero norm. Square integrability requires

$$\|f\|^2 = \int_0^{2\pi} |f(x)|^2\, dx < \infty.$$

The functions $1, \cos nx, \sin nx$, for positive integer n, are mutually orthogonal in the appropriate pairs. For example,

$$\int_0^{2\pi} \sin x \cos x\, dx = 0.$$

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Functions as vectors (continued)

An infinite trigonometric system requires completeness and a convergence notion to be a Hilbert-space basis. Pointwise convergence requires additional conditions; L^2 convergence is the natural projection framework.

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Fourier coefficients as projections

Begin with a finite truncation.

$$f_N(x) = a_0 + \sum_{n=1}^{N} (a_n \cos nx + b_n \sin nx).$$

Take an inner product with each basis function; orthogonality removes the other terms.

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, dx,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx\, dx,$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx\, dx.$$

These are least-squares projection coefficients in the selected trigonometric subspace. Under 2π-periodic extension, odd functions have zero cosine coefficients and even functions have zero sine coefficients. This parity is about the origin, not merely a shape seen on $[0, 2\pi]$.

Fourier coefficients as projections (continued)

Let $e_N = f - f_N$. Since the residual is orthogonal to every retained basis function,

$$\|f\|^2 = \|f_N\|^2 + \|e_N\|^2.$$

adding an independent orthogonal mode transfers that mode's energy from the residual to the approximation. A zero coefficient leaves the result unchanged. This is a finite orthogonal-projection energy decomposition. The infinite Parseval equality additionally requires completeness of the orthogonal system; convergence in L^2 alone does not imply pointwise or uniform convergence. Rates depend on smoothness, jumps, and boundary behavior; jumps produce Gibbs oscillations near the discontinuity.

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Homogeneous coordinates

An affine map on $\mathbb{R}^3$,

$$y = Rx + t$$

becomes a linear matrix map in homogeneous coordinates.

$$\begin{bmatrix} y \\ 1 \end{bmatrix} = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}.$$

Translation:

$$T(t) = \begin{bmatrix} I & t \\ 0 & 1 \end{bmatrix}.$$

Scaling:

$$S(s_x, s_y, s_z) = \begin{bmatrix} \mathrm{diag}(s_x, s_y, s_z) & 0 \\ 0 & 1 \end{bmatrix}.$$

Homogeneous tuples

$$(x_0, x_1, x_2, w) \sim (\lambda x_0, \lambda x_1, \lambda x_2, \lambda w)$$

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Homogeneous coordinates (continued)

represent the same projective point when $\lambda \neq 0$ and the tuple is nonzero. Finite points normalize to $w = 1$. Directions use $w = 0$, so translations leave them unchanged.

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Composing graphics transformations

To rotate around p, translate p to the origin, rotate, then translate back. With column vectors,

$$T(p)R_\theta T(-p)x.$$

the rightmost transformation acts first.

For a plane through the origin with unit normal n, projection and reflection are

$$P = I - nn^T, \qquad M = I - 2nn^T.$$

For an offset plane, translate it to the origin and back. Perspective uses a nonzero homogeneous component followed by division. With the eye at the origin and image plane $z = d > 0$,

$$P_{\text{persp}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/d & 0 \end{bmatrix}.$$

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Composing graphics transformations (continued)

a point $[x, y, z, 1]^T$ becomes $[x, y, z, z/d]^T$. For $z \neq 0$, division by z/d yields $(dx/z, dy/z, d)$ on the image plane; this camera model normally retains $z > 0$. The matrix has rank 3: it defines central perspective projection, not an invertible projective transformation. At $z = 0$ division is undefined, and the origin maps to the zero vector, which is not a projective point. Clipping and depth conventions depend on the graphics API.

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Finite fields and the Hill cipher

For prime p, the field

$$\mathbb{F}_p = \{0, 1, \ldots, p-1\},$$

uses addition and multiplication modulo p. Every nonzero element has a reciprocal. A square key matrix is invertible exactly when

$$\det K \not\equiv 0 \pmod{p}.$$

Encode:

$$y = Kx \pmod{p}.$$

Decode:

$$x = K^{-1}y \pmod{p}.$$

This defines a Hill cipher. For $p = 29$ and

$$K = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix},$$

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Finite fields and the Hill cipher (continued)

the determinant is $-1 \equiv 28 \pmod{29}$, so the key is invertible. An affine version adds a fixed message-space translation.

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Chapter review

- Differences and positive conductances produce A^TCA; full-network injections must balance.
- Markov convergence requires the stated irreducibility and aperiodicity conditions; periodic chains can have stationary initial states.
- A zero ratio is a valid simplex step; a dual certificate requires feasibility.
- Orthogonal projections, homogeneous coordinates, and perspective division have distinct assumptions.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 10.

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